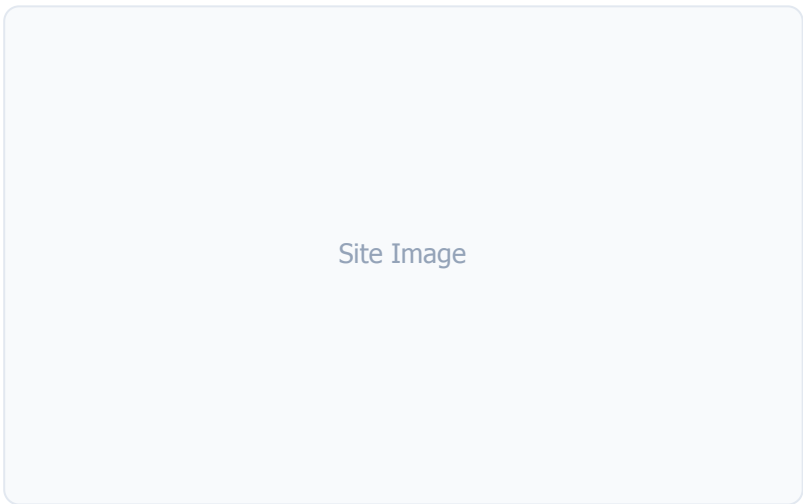


INITIAL STATIC VERTICAL PILE LOAD TEST ON 900MM DIA PILE

FOR BDD Chawls Redevelopment Project, Worli, Mumbai
AT Building-3 Wing B

(TEST PILE TP-01)



Site Image

Client:	Tata Project
Contractor:	NA
Test Date:	08 December 2025
Report No:	IVPLT-001

Test conducted as per IS 2911 (Part 4) - 2013

CONTENTS

1.0 General	Page 3
.....	
2.0 Scope of Work	Page 4
.....	
3.0 Methodology	Page 5
.....	
4.0 Results	Page 6
.....	
5.0 Readings and Graph	Page 7
.....	
6.0 Conclusion	Page 8
.....	
7.0 Load Test Data	Page 9
.....	

08 December 2025

IVPLT Report - TP-01

1.0 General

1.1 Tata Project decided to carry out static pile testing work on 900mm diameter pile to estimate load carrying capacity in vertical direction and settlement. This is the Initial Vertical Pile Load Test at Building-3 Wing B.

1.2 This report covers data for one vertical pile load test. This report covers calculation of safe load capacity for pile based on data collected during fieldwork.

1.3 The following codes of practices have been adopted:

- IS 2911 (Part 4) - 2013 "Code of Practice for Design and Construction of Pile Foundations - Load Tests on Piles"
- IS 14593 - 1998 (Reaffirmed 2003) "Design and Construction of Bored Cast-in-Situ Piles Founded on Rocks – Guidelines"

08 December 2025

IVPLT Report - TP-01

2.0 Scope of Work

Pile details are tabulated as below.

2.1 Pile Details for Initial Pile (Vertical Load Test)

Location	Building-3 Wing B
Pile ID	TP-01
Pile Diameter	900 mm
Pile Depth	11.51 m
Concrete Grade	M40
Maximum Vertical Safe Capacity	420 MT

2.2 Vertical Test Load for Initial Pile

The design vertical load on the pile is **420 MT**.

The pile is required to be tested to a load of **1050 MT** ($2.5 \times$ design load).

2.3 Equipment Details

Hydraulic Jack	As per site specifications
Ram Area	2551 cm²
Dial Gauge Least Count	0.01 mm

3.0 Methodology

The Initial Vertical Pile Load Test was conducted in accordance with IS 2911 (Part 4) - 2013 "Code of Practice for Design and Construction of Pile Foundations - Load Test on Piles".

3.1 Test Setup

A hydraulic jack of adequate capacity was used to apply the load, reacting against a sturdy reaction frame. Four dial gauges (0.01 mm least count) were fixed on the pile head to record settlements.

3.2 Test Procedure

1. Load was applied in increments of approximately 20% of design load (84.0 MT) up to the test load of 1050 MT.
2. Each load increment was maintained until the rate of settlement was less than 0.1mm/hour, or for a minimum of 1 hour.
3. Maximum load was maintained for 24 hours as per IS 2911 requirements for initial tests.
4. Load was released in the same increments as the loading phase.
5. Settlement readings were recorded at each load increment using four dial gauges positioned at 90° intervals.

3.3 Acceptance Criteria (IS 2911 Part 4)

As per IS 2911 (Part 4) - 2013, the acceptance criteria for Initial Vertical Pile Load Test are:

1. Net settlement at design load shall not exceed 12mm or 2% of pile diameter (18.0mm), whichever is less.
2. Safe load shall be taken as two-thirds of the load at which total settlement equals 12mm.
3. Safe load shall not exceed half the load at which total settlement equals 10% of pile diameter (90.0mm).

08 December 2025

IVPLT Report - TP-01

4.0 Results

4.1 Test Results Summary

Maximum Settlement Recorded	7.22 mm
Elastic Rebound	3.08 mm
Net Settlement (Max - Rebound)	4.14 mm
Settlement Limit (IS 2911)	12 mm
Safe Load Adopted	420.0 MT
Governing Criterion	DESIGN
Test Status	PASSED

4.2 Assessment

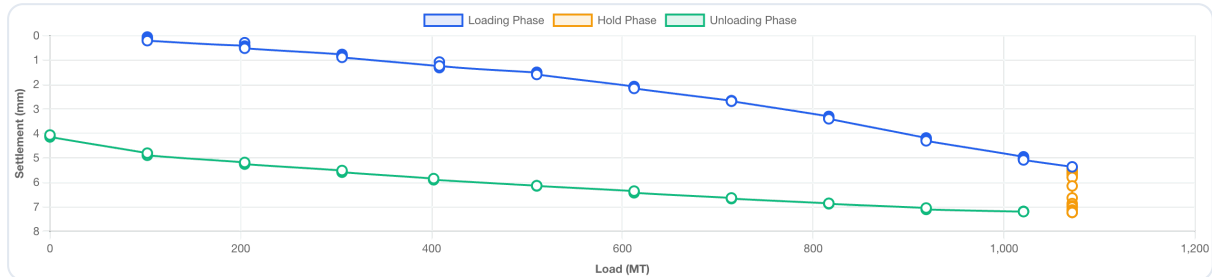
The net settlement of 4.14 mm is within the permissible limit of 12 mm as per IS 2911 (Part 4) - 2013.

The safe load adopted for the pile is **420.0 MT** based on the design criterion.

08 December 2025

IVPLT Report - TP-01

5.0 Load vs Settlement Curve



TEST LOAD

1050.0

MT

MAX SETTLEMENT

7.22

mm

NET SETTLEMENT

4.14

mm

TEST PASSED ✓

Net settlement 4.14mm is within the 12mm limit (IS 2911 Part 4)

08 December 2025

IVPLT Report - TP-01

6.0 Conclusion

Based on the Initial Vertical Pile Load Test conducted on pile TP-01 at Building-3 Wing B, the following observations and conclusions are drawn:

1. The pile was loaded up to 1050.0 MT (2.5 times the design load of 420 MT) as per IS 2911 (Part 4) - 2013.
2. The maximum settlement recorded at test load was 7.22 mm.
3. After complete unloading, the elastic rebound was 3.08 mm.
4. The net settlement (residual) is 4.14 mm, which is within the permissible limit of 12 mm.
5. The safe load adopted for the pile is 420.0 MT based on the design criterion.
6. **The pile HAS PASSED the Initial Vertical Pile Load Test as per IS 2911 (Part 4) - 2013.**

Therefore, the design safe load of 420 MT can be adopted as the working load capacity for this pile.

08 December 2025

IVPLT Report - TP-01

7.0 Load Test Data

S.NO	PHASE	LOAD (MT)	SETTLEMENT (MM)	SIGNATURE
1	LOADING	0.00	0.00	
2	LOADING	102.04	0.06	
3	LOADING	102.04	0.13	
4	LOADING	102.04	0.17	
5	LOADING	102.04	0.19	
6	LOADING	102.04	0.21	
7	LOADING	204.08	0.41	
8	LOADING	204.08	0.29	
9	LOADING	204.08	0.43	
10	LOADING	204.08	0.47	
11	LOADING	204.08	0.52	
12	LOADING	306.12	0.77	
13	LOADING	306.12	0.81	
14	LOADING	306.12	0.83	
15	LOADING	306.12	0.86	
16	LOADING	306.12	0.89	
17	LOADING	408.16	1.25	
18	LOADING	408.16	1.28	
19	LOADING	408.16	1.31	
20	LOADING	408.16	1.08	
21	LOADING	408.16	1.24	
22	LOADING	510.20	1.51	
23	LOADING	510.20	1.50	
24	LOADING	510.20	1.53	
25	LOADING	510.20	1.56	

S.NO	PHASE	LOAD (MT)	SETTLEMENT (MM)	SIGNATURE
26	LOADING	510.20	1.59	
27	LOADING	612.24	2.08	
28	LOADING	612.24	2.11	
29	LOADING	612.24	2.13	
30	LOADING	612.24	2.13	
31	LOADING	612.24	2.16	
32	LOADING	714.28	2.65	
33	LOADING	714.28	2.65	
34	LOADING	714.28	2.67	
35	LOADING	714.28	2.68	
36	LOADING	714.28	2.69	
37	LOADING	816.32	3.30	
38	LOADING	816.32	3.33	
39	LOADING	816.32	3.35	
40	LOADING	816.32	3.38	
41	LOADING	816.32	3.40	
42	LOADING	918.36	4.18	
43	LOADING	918.36	4.20	
44	LOADING	918.36	4.23	
45	LOADING	918.36	4.27	
46	LOADING	918.36	4.30	
47	LOADING	1020.40	4.95	
48	LOADING	1020.40	5.00	
49	LOADING	1020.40	5.03	
50	LOADING	1020.40	5.06	
51	LOADING	1020.40	5.08	
52	LOADING	1071.42	5.36	
53	HOLD	1071.42	5.40	

S.NO	PHASE	LOAD (MT)	SETTLEMENT (MM)	SIGNATURE
54	HOLD	1071.42	5.44	
55	HOLD	1071.42	5.50	
56	HOLD	1071.42	5.53	
57	HOLD	1071.42	5.58	
58	HOLD	1071.42	5.60	
59	HOLD	1071.42	5.62	
60	HOLD	1071.42	5.66	
61	HOLD	1071.42	5.67	
62	HOLD	1071.42	5.68	
63	HOLD	1071.42	5.74	
64	HOLD	1071.42	5.79	
65	HOLD	1071.42	6.14	
66	HOLD	1071.42	6.63	
67	HOLD	1071.42	6.86	
68	HOLD	1071.42	6.95	
69	HOLD	1071.42	7.00	
70	HOLD	1071.42	7.12	
71	HOLD	1071.42	7.17	
72	HOLD	1071.42	7.19	
73	HOLD	1071.42	7.21	
74	HOLD	1071.42	7.22	
75	HOLD	1071.42	7.22	
76	HOLD	1071.42	7.22	
77	UNLOADING	1020.40	7.18	
78	UNLOADING	1020.40	7.18	
79	UNLOADING	1020.40	7.18	
80	UNLOADING	918.36	7.08	
81	UNLOADING	918.36	7.06	

S.NO	PHASE	LOAD (MT)	SETTLEMENT (MM)	SIGNATURE
82	UNLOADING	918.36	7.04	
83	UNLOADING	816.32	6.87	
84	UNLOADING	816.32	6.84	
85	UNLOADING	816.32	6.84	
86	UNLOADING	714.24	6.66	
87	UNLOADING	714.24	6.64	
88	UNLOADING	714.24	6.63	
89	UNLOADING	612.24	6.41	
90	UNLOADING	612.24	6.37	
91	UNLOADING	612.24	6.34	
92	UNLOADING	510.20	6.14	
93	UNLOADING	510.20	6.13	
94	UNLOADING	510.20	6.13	
95	UNLOADING	402.16	5.89	
96	UNLOADING	402.16	5.87	
97	UNLOADING	402.16	5.84	
98	UNLOADING	306.12	5.58	
99	UNLOADING	306.12	5.54	
100	UNLOADING	306.12	5.51	
101	UNLOADING	204.08	5.25	
102	UNLOADING	204.08	5.21	
103	UNLOADING	204.08	5.18	
104	UNLOADING	102.04	4.89	
105	UNLOADING	102.04	4.82	
106	UNLOADING	102.04	4.80	
107	UNLOADING	0.00	4.14	
108	UNLOADING	0.00	4.11	
109	UNLOADING	0.00	4.06	

08 December 2025

IVPLT Report - TP-01

Prepared by:

Checked by:

Site Engineer

Quality Engineer

Reviewed by:

Approved by:

Project Manager

Client Representative